

Chapter 2 In Class Exercises

Processes Threads Synchronization and Scheduling

Name		Date	
Group		Class	

Purpose Use the process and thread models from Chapter 2 to trace system behavior, identify synchronization problems, and compare scheduling results.

Suggested format Work in groups of two or three. Complete the trace or calculation first, then agree on one concise explanation. Suggested time is 60 to 75 minutes.

Exercise 1 Process States

Directions For each event, write the process state before the event, the state after the event, and the reason for the transition. Use Running, Ready, Blocked, or Terminated.

Event	Before	After	Reason
The scheduler selects P1 from the ready queue.			
P1 requests data from the disk.			
The disk finishes P1's request.			
A timer interrupt occurs while P2 uses the CPU.			
P2 finishes normally.			

1A. Can a blocked process move directly to Running in the three-state model? Explain in one sentence.

Exercise 2 Processes and Threads

Directions Mark each item Shared if threads in the same process normally share it, or Private if each thread normally has its own copy.

Item	Shared or Private	Why
Address space		
Global variables		
Open files		
Program counter		
CPU registers		
Stack		
Thread state		

2A. A word processor uses one thread for the user interface, one for spell checking, and one for saving. Give one benefit and one risk of this design.

Exercise 3 Race Condition Trace

Directions Two threads share counter = 10. Each executes counter = counter + 1. The read and write are separate machine actions.

Step	Action	T1 local value	T2 local value	Shared counter
1	T1 reads counter		-	
2	T2 reads counter			
3	T1 adds 1 and writes			
4	T2 adds 1 and writes			

3A. What final value does this interleaving produce? What value should correct execution produce?

3B. Underline or describe the critical region that must execute atomically.

Exercise 4 Producer Consumer Semaphores

Directions A bounded buffer has 4 slots. Initially empty = 4, full = 0, and mutex = 1. Fill in the semaphore values after each completed operation. If an operation blocks, write BLOCKED and leave later cells blank.

Step	Operation	empty	full	mutex	Blocks
Start	Initial values	4	0	1	No
1	Producer down(empty)				
2	Producer down(mutex)				
3	Producer inserts item and up(mutex)				
4	Producer up(full)				
5	Consumer down(full)				
6	Consumer down(mutex)				
7	Consumer removes item and up(mutex)				
8	Consumer up(empty)				

4A. After four producers insert four items and no consumer runs, what are empty and full?

4B. What happens when a fifth producer executes down(empty)? Why is mutex alone not enough for this buffer?

Exercise 5 Barrier Timing

Directions Four threads finish phase 1 at the times shown. A barrier prevents any thread from starting phase 2 until all four arrive.

Thread	Arrival at barrier	Release time	Waiting time
T1	5 ms		
T2	8 ms		
T3	3 ms		
T4	11 ms		

5A. At what time may phase 2 begin? Calculate the average waiting time at the barrier.

Exercise 6 Multiprogramming and CPU Utilization

Directions Assume each process waits for I/O 60 percent of the time. Use $U = 1 - p^n$, where p is the probability that one process waits for I/O and n is the number of processes in memory.

Processes n	Calculation	CPU utilization
1	$1 - 0.60^1$	
2	$1 - 0.60^2$	
3	$1 - 0.60^3$	
4	$1 - 0.60^4$	
5	$1 - 0.60^5$	

6A. What is the smallest n that gives at least 90 percent CPU utilization?

6B. Why does adding processes improve utilization in this model? State one limitation of the model.

Exercise 7 Batch Scheduling

Directions All three processes arrive at time 0. Compare First Come First Served and Shortest Job First. Turnaround time = completion time - arrival time. Waiting time = turnaround time - burst time.

Process	Burst time
P1	6 ms

Process	Burst time
P2	2 ms
P3	4 ms

7A. Draw the FCFS Gantt chart using the order P1, P2, P3.

FCFS	Completion	Turnaround	Waiting
P1			
P2			
P3			
Average			

7B. Draw the nonpreemptive SJF Gantt chart, then complete the table.

SJF	Completion	Turnaround	Waiting
P1			
P2			
P3			
Average			

Exercise 8 Round Robin Scheduling

Directions Use the same processes from Exercise 7 and a 2 ms time quantum. All processes arrive at time 0 in the order P1, P2, P3.

8A. Draw the Round Robin Gantt chart.

Round Robin	Completion	Turnaround	Waiting
P1			
P2			
P3			
Average			

8B. Which policy gives the shortest average turnaround time for this workload: FCFS, SJF, or Round Robin?

8C. Why might an interactive system still prefer Round Robin? What changes when the time quantum becomes extremely small?

Exercise 9 Priority Inversion

Directions A low priority task L holds a mutex. A high priority task H becomes ready and needs the same mutex, so H blocks. A medium priority task M does not need the mutex and preempts L.

9A. Explain why this situation is called priority inversion.

9B. Under priority inheritance, whose priority does L temporarily receive? When should L return to its original priority?

Exit Discussion

10. Choose one server design from the slides: single threaded, multithreaded, or event driven. Give one workload where it would be a reasonable choice and one tradeoff.